

Specifying Operational Design Domain in Autonomous Driving for Comprehensive Data Evaluation

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Abstract—Operational Design Domain (ODD) attributes define the environmental conditions under which Automated Driving Systems (ADS) can safely operate. These attributes include factors such as road and lighting conditions, as well as infrastructure elements, such as lane markings and road conditions. However, existing ODD definitions are often ambiguous and lack specificity, making it challenging to validate their presence in datasets.

The absence of precise ODD definitions and robust validation mechanisms poses significant challenges, as it remains unclear whether ADS training and testing datasets adequately represent real-world operating conditions. This gap introduces risks that could compromise the safe deployment of ADS in diverse environments.

To address this issue, we introduce FODSE (Framing ODDs as Domain Specifications for Evaluation), a semi-automated AI-powered approach that refines ODD attributes into structured, context-aware domain specifications and systematically evaluates their presence in datasets. FODSE leverages Retrieval-Augmented Generation (RAG), multimodal AI, and prompt learning to enhance specification clarity and dataset completeness.

Experimental evaluation on two commonly adopted datasets in ADS demonstrates that FODSE significantly improves dataset validation accuracy, achieving up to 96.8% classification accuracy for an extended set of lane marking variants and 97.8% for roadway users—two key ODD attributes. Expert assessments confirm that FODSE effectively reduces ambiguity and enhances contextual adaptability, reinforcing its potential to improve dataset integrity and ensure safer, more reliable ADS training and validation.

I. INTRODUCTION

In the context of ADS features, such as automatic lane changes and self-parking, it is customary to define the Operational Design Domain (ODD) within which a feature is expected to operate. ODDs and requirements are closely linked. The ODD defines a set of constrained scenarios in which the feature must function, thereby establishing the operational scope [15], [21]. In turn, the requirements specify the expected behavior of the feature within the scope defined by its ODD. For example, if the ODD permits operation in low-light conditions, the requirements must specify the sensor capabilities necessary for nighttime perception. While ODDs serve as the foundation for requirements engineering, the rigorous practices associated with the latter have not been

consistently applied to ODD descriptions. As a result, ODDs are often broad and less formally defined.

A key challenge in ensuring the safety of ADS features lies in the accurate definition and validation of their ODD boundaries [7], [15], [38]. However, in practice, documentation related to ODD often oversimplifies these boundaries into lists of attributes or objects (e.g., road types, weather conditions, traffic environments), failing to capture the contextual nuances necessary for safe and effective operation [1]. ODD attributes in ADS, such as roadway types, geographic features, speed ranges, and weather conditions, lack specificity and can vary significantly based on geographical or contextual differences [37]. Unlike traditional domain features in RE, which are usually well-structured and explicitly tied to system requirements, ODD attributes are inherently high-level and abstract, making it difficult to establish precise boundaries.

The challenge is further compounded by the presence of machine learning components in many ADS features, which amplifies the need for precise ODD specifications. Conventional software validation relies on traceability from requirements specifications to system-level tests, ensuring that each component meets predefined specifications. In contrast, machine learning systems depend on training data rather than structured design processes, making validation inherently more complex. As a result, validation efforts must focus on ensuring that both training and testing datasets comprehensively represent all relevant operational conditions, guaranteeing robust and reliable system performance across diverse scenarios [33].

Hence, aside from specifying the ODD attributes, ensuring that vision-based training and testing data (e.g., image and video frames) contain these conditions presents a significant challenge. Accurately mapping these conditions to pixel-level information in image and video data is a non-trivial but essential task for evaluating the completeness of the data and the overall performance of automated driving systems.

The research in this area is limited. For instance, Czarnecki et al proposed a manually-built taxonomy for the ODD of ADS based on existing standards [7], while another study suggested an architectural design to extract ODDs from task scenarios and requirements [9]. While these studies focus on defining

the ODD attributes, this paper advances the field by proposing a semi-automated approach, called FODSE (Framing ODDs as Domain Specifications for Evaluation), for ODD specifications and the evaluation of their presence in data, by leveraging advanced AI technologies—including generative AI, multimodal models, and prompt engineering.

FODSE translates the inherently vague and ambiguous ODD attributes into context-sensitive domain specifications that resemble structured RE features. These specifications are then refined by incorporating subjective nuances and regional variations in terminology to be used to evaluate the training and testing data. Therefore, this paper’s objectives are twofold: (1) To extract contextualized domain specifications from ODDs and formalize them into a knowledge graph. (2) To evaluate the extracted specifications in a dataset to identify gaps in data that can impact ADS training and overall performance.

FODSE is built upon two integrated components: the *Specification Augmentation Engine* and the *Dataset Evaluation Engine*. The *Specification Augmentation Engine* transforms often ambiguous ODD attributes into structured, context-specific specifications, by leveraging Retrieval-Augmented Generation (RAG) and Knowledge Graph (KG) construction to ground these specifications. Building on these refined specifications, the latter engine analyzes visual datasets to identify coverage gaps with respect to the defined ODD conditions. It systematically maps structured ODD attributes, such as roadway users, to image-level annotations. To improve coverage detection and semantic alignment, FODSE integrates multimodal models (MMs) augmented with prompt learning techniques, allowing it to more effectively recognize underrepresented operational conditions. This enables requirements engineers to assess whether datasets used for training and testing adequately represent the full operational scope of an ADS feature.

By reframing informal ODD attributes as structured domain-level requirements, FODSE directly supports foundational principles in RE, including traceability, ambiguity resolution, and specification completeness. The specifications generated by FODSE—covering functional expectations, non-functional constraints, and contextual adaptability—are intentionally structured to align with traditional RE artifacts such as domain models, requirements documents, and safety case evidence. These artifacts facilitate end-to-end traceability: from high-level ODD constraints, to dataset composition, and ultimately to observable system behavior. This alignment not only enhances verifiability and auditability but also promotes a requirements-driven development paradigm for AI-enabled autonomous systems. In this way, FODSE bridges a critical gap between RE theory and the practical demands of perception-driven, data-dependent ADS development.

To evaluate our approach, we applied FODSE to two ODD categories—lane markings and roadway users—assessing its ability to translate ODD attributes into structured requirement specifications and to analyze image datasets effectively. Expert evaluations measured the specificity, ambiguity resolution, and contextual adaptability of the generated specifications, highlighting FODSE’s ability to improve ODD structuring

while identifying areas for refinement, such as incorporating environmental factors like weather and visibility.

We utilized two widely used large-scale autonomous driving datasets, BDD100K (Berkeley DeepDrive) and TuSimple, to evaluate our approach. BDD100K is a key resource for evaluating perception models in diverse driving environments. Similarly, TuSimple is a dataset specifically designed for lane detection, containing highway road images with annotated lane markings, commonly used to benchmark lane detection algorithms in highway scenarios.

Our experimental results on these datasets, using CLIP, BLIP2, and a prompt-learned version of CLIP within FODSE, demonstrated that prompt learning significantly improved classification accuracy of FODSE, reducing false positives and increasing F1-scores across all tested classes. These findings highlight the importance of prompt optimization and dataset augmentation in improving dataset reliability, ensuring that ADS models are trained on well-defined, context-aware specifications that align with real-world operational conditions.

Furthermore, our evaluation revealed that multiple ODD attribute classes identified by FODSE were missing from these commonly used datasets, emphasizing FODSE’s effectiveness in systematically detecting dataset gaps. This underscores its value in ensuring comprehensive ODD coverage and improving dataset completeness for autonomous system development. However, it also highlights a limitation in our evaluation, as we were constrained to assessing only the attributes with existing ground truth labels. These findings further validate the need for dataset augmentation and more representative benchmarks to enhance the robustness of ADS training and evaluation.

This work was conducted in collaboration with our industrial partners at General Motors R&D, leveraging their expertise and insights to address real-world challenges in ADS development.

II. ODD DEFINITION AND CHALLENGES

TABLE I
EXAMPLES OF INITIAL ODD ATTRIBUTES FOR ADS FEATURES

Category	Attributes
Traffic Conditions	traffic density, speed limits.
Weather Conditions	Rain, snow, fog, clear skies.
Geographical Conditions	Urban, suburban, rural areas.
Lane Markings	Continuous, dashed lines, lane width.
Lighting Conditions	Daytime, nighttime, varying ambient light levels.
Road Type	Highways, arterial roads, local streets.
Roadway Users	Pedestrians, cyclists, motorcyclists, vehicles.
Density and Flow	High pedestrian areas, light/heavy traffic.
User Attributes	Speed, move direction, traffic signal compliance.

As displayed in Table I, ADS ODDs encompass various categories, such as *weather conditions*, which include attributes, such as *rain* and *snow* and may involve environmental, geographical, and time-based elements. Automated systems should adapt their functionality according to the ODD

conditions. For instance, if an autonomous vehicle detects *heavy rain*, which might obstruct its front-facing radar, it may deactivate certain features like adaptive cruise control to ensure safety. The ODD attributes are often too general and abstract to serve as effective guidelines for developing AI-enabled systems (AIS), such as ADS. One primary issue is the lack of specificity. For instance, the attribute *traffic density* can vary widely without clear definitions or quantitative measures (e.g., vehicles per kilometer). Another challenge is the ambiguity in terms like *urban*, *suburban*, *rural*, or *weather conditions* like *rain* and *fog*, which are subjective and can differ significantly depending on the region or context. For example, *light snow* in a *rural area* can create entirely different road conditions than the same weather in an urban environment.

A key challenge after framing ODDs as domain specifications is validating and ensuring their representation in datasets. Even if ODDs are translated into detailed specifications, the effectiveness of an ADS depends on whether these specifications are adequately reflected in the training data. If certain conditions are missing or underrepresented, the AIS (e.g., ADS) may struggle to handle them in real-world scenarios. Therefore, it is critical to verify that the dataset comprehensively covers the necessary conditions and environmental variations, as gaps in representation could negatively impact system performance and safety.

III. RELATED WORKS

While formal ODD definitions exist in standards such as SAE J3016 [14] and PAS 1883 [44], their translation into practical artifacts remains limited. Academic proposals—including human-readable formats [15], SQL encodings [38], and logic-based formulations for driving simulators [43]—primarily focus on enumerating attributes or enabling test generation. However, they do not provide mechanisms for interpreting, enriching, or validating these attributes in dataset contexts. This gap between high-level ODD declarations and dataset-level implementation motivates our specification- and evaluation-driven approach.

ODD-inspired abstractions have also been applied in rail and maritime domains, but with a focus on system-level properties rather than data-centric specifications. Tonk and Boussif [45] reinterpret ODD as “operational envelope” and “operational context,” yet do not offer methods for dataset evaluation. Meng et al. [27] and Cuartas et al. [6] use UPPAAL to check safety properties in autonomous systems, but their scope remains model-centric, lacking pathways for coverage analysis or augmentation—limitations that FODSE directly addresses.

Some efforts have linked ODD to software engineering practices, though often without supporting specification formalization or dataset assessment. Lee [22] and Cho [5] propose hazard-aware ODD delimitation, but do not yield machine-readable representations for validation workflows. Gyllenhammar et al. [12] connect ODD to safety case structure, yet do not operationalize this link for perception data. FODSE extends these works by reframing high-level assumptions as domain-grounded specifications for validating dataset structure and

model behavior—applying domain engineering principles [4] to data-driven ADS validation.

Formal verification tools such as Z3 [8] and CVC5 [3] have long supported system-level validation through satisfiability and model checking techniques. However, these approaches presuppose fully formalized specifications and are ill-suited to contexts involving partial, evolving, or data-centric requirements. In contrast, FODSE focuses on enabling specification augmentation and dataset evaluation, which calls for complementary techniques grounded in recent advances in machine learning and language modeling.

Recent work in RE increasingly integrates large language models (LLMs) to support evolving, complex requirements. Research such as [10] surveys LLM applications across RE tasks. Prompt-based approaches [42], along with safety-oriented studies [26], [29], address elicitation and inconsistency detection. FODSE builds on these efforts by transforming raw LLM output into structured, traceable specifications tailored for dataset validation.

Retrieval-augmented generation (RAG) has proven effective for knowledge-intensive reasoning [11], domain-specific tuning [48], and retrieval-aware QA [40], though applications in RE remain sparse. Arora et al. [2] generate test scenarios from NL requirements using RAG. FODSE continues this line by using RAG to contextualize vague ODD attributes via expert and regulatory sources—mitigating ambiguity and enhancing specification clarity.

The integration of LLMs with KGs holds potential for improving ADS. Prior research, such as [47], demonstrates how LLMs can enhance the accuracy of creating KGs from unstructured text, highlighting the role of advanced models in semantic reasoning. Studies like [32] explore the complementary strengths of LLMs and KGs, proposing frameworks for their unification to enhance inference and interpretability. Additionally, [53] introduces methods to incorporate structural information into LLMs for better KG completion. FODSE pragmatically applies these ideas by structuring LLM-generated RAG outputs into augmented KGs for downstream validation. This approach enables traceable, semantically grounded specifications aligned with multimodal perception data.

Prompt engineering plays a key role in optimizing MM model performance for ADS. Prior work [41] highlights ChatGPT’s sensitivity to prompt design in data analysis, while studies such as [16], [20] explore methods to refine prompts for task-specific improvements. Other works [18], [35], [55] show how prompt tuning and context control enhance vision-language models in zero- and few-shot settings. While these efforts center on prompt design itself, FODSE adopts a practical use of prompt engineering via PromptSRC, tailoring domain-specific prompts to support accurate classification and validation of structured ODD specifications.

IV. PROPOSED FRAMEWORK

The challenges outlined in Section II highlight the need for a more structured approach to utilizing ODDs as practical guidelines to develop ADS. To address this, we propose a

framework that establishes more precise constraints to overcome the lack of specificity present in current ODD definitions, reduce their ambiguities, and better capture contextual differences through domain-specific clarifications, ensuring that ODDs are translated into more actionable guidelines for AI system development. By reframing the ODD attributes as extended domain specifications, our framework facilitates the evaluation of ADS datasets against these criteria, ensuring that they encompass the necessary environmental and contextual variations. This enables AI systems to train on diverse data and effectively manage the full spectrum of domain specifications.

FODSE, as shown in Figure 1, is composed of two primary components: **Specification Augmentation** and **Dataset Evaluation** engines. The former focuses on converting ODDs into context-specific specifications and further augmenting them with domain knowledge. The latter utilizes the output of the first engine to assess ADS datasets against these refined specifications.

A. Specification Augmentation Engine

The extraction engine operates by leveraging a combination of Retrieval-Augmented Generation (RAG) (Figure 1-①) and Knowledge Graph (KG) (Figure 1-②) construction to extend ODDs into more detailed domain specifications.

1) *Retrieval-Augmented Generation (RAG)*: The concept of RAG was initially proposed in [23] with the original use case of enhancing generative AI models by linking them to external and up-to-date resources. In a RAG setup, the model improves its generated responses by retrieving and processing additional relevant documents, which are then used to refine its output. As such, the models generative capabilities extend beyond traditional information retrieval, as it combines both the retrieved content and the model’s prior knowledge, enabling it to produce contextually precise answers that are more accurate and relevant to the task at hand.

For this purpose, we utilized GPT-4 Turbo to develop an RAG system, where the model dynamically integrated retrieved domain-specific documents with its prior knowledge to generate context-aware and precise specifications. For this study, we utilized publicly available documents as a proof of concept to demonstrate our approach since industry-specific documents contained proprietary and confidential information requiring restricted access. However, we believe our framework is highly adaptable, allowing seamless integration with domain-specific sources as needed for various applications. The relevant information was retrieved from external regulatory guidelines and standards documents, including *Signalized Intersections: Informational Guide* [36] and *Verification, Refinement, and Applicability of Long-Term Pavement Performance Vehicle Classification Rules* [13], *Agricultural Equipment on Public Roads* [30], and *Pedestrian Facilities Users Guide: Providing Safety and Mobility* [51] (full list of documents is available in our online repository).

The RAG setup embedded this domain knowledge into numerical vectors and stored in a database to be passed to the generative model to produce detailed specifications. The

integration of RAG in our framework enabled the generative model to produce context-sensitive responses by grounding its output in real-world, region-specific data.

We used RAG in FODSE because it allowed us to directly address key challenges related to ODDs, including their lack of specificity, ambiguity, and contextual differences. By referencing domain-specific documents, RAG helped generate more accurate and detailed domain specifications that were aligned with the regional regulations and standards that we provided. This capability ensured that our ODD definitions were precise and context-aware, allowing us to move beyond general and abstract descriptions. For instance, after retrieving foundational data on traffic regulations or environmental conditions, RAG formulated specific thresholds for weather conditions or traffic flow based on the context-related standards provided. Meanwhile, the generic definitions of these conditions were derived from RAG’s pre-trained knowledge, ensuring a balance between domain specificity and broader applicability.

This retrieval-based generative process ensured that the specifications were both accurate and contextually adapted to the operational standards provided to the model. For example, RAG retrieved relevant regulatory data to refine ambiguous terms such as *traffic density*, translating them into quantifiable metrics like *vehicles per kilometer*. Additionally, it tailored requirements with contextual adaptability, aligning them with specific standards and guidelines by defining precise values, such as the number of vehicles per kilometer for different traffic density conditions. In this case, the metric was adjusted based on peak versus non-peak traffic hours, as discussed in the documents. Similarly, it refined ambiguous terms like *urban* into quantifiable attributes, such as population density or road infrastructure. By capturing contextual variations—such as differing *road conditions* across regions—RAG produces tailored domain specifications, enhancing the clarity, precision, and applicability of ODDs.

While RAG can function independently, we adopted a human-in-the-loop approach to verify outputs, ensuring the generated domain specifications were retrieved from the provided regulations and standards. This combination of human oversight and automated generation ensured that the outputs were both precise and contextually accurate.

Here, we provide two examples using the ODD attribute for *lane markings* and *roadway users*. The process is the same for both ODD categories. The first human-in-the-loop submitted a query specifying the ODD attribute requiring further specification, for example, “What are the different types of [lane markings] and their meanings in different traffic environments?” This step initiates the retrieval process, narrowing down the area of focus within the broader set of ODDs.

The engine retrieved context-relevant information from the vectorized database containing regulatory documents, traffic guidelines, and road safety manuals using semantic search. These sources provide detailed definitions of ODD attributes, such as lane markings, roadway users, traffic density, and road conditions. The engine processes the query and returns

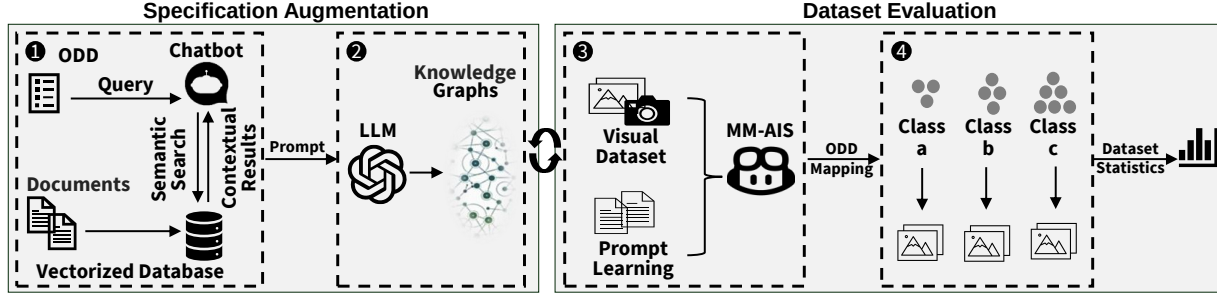


Fig. 1. An Overview of FODSE primary Components.

relevant passages, such as:

```
Query: "What are the different types of lane markings and
       their meanings?"
Documents: Sources include traffic regulations (MUTCD, SAE
       J3016) with the following content:
       - Solid White Line: "Indicates the edge of the roadway
       and separates traffic moving in the same
       direction."
       - Broken White Line: "Separates lanes of traffic moving
       in the same direction; lane changes are permitted."
       - Double Yellow Line: "Indicates a no-passing zone for
       traffic in both directions."
# Retrieved content continues...
```

For more details, refer to our repository, where we provide an in-depth overview of the retrieval process and the augmented domain specifications¹. The human-in-loop then submits an augmented prompt to RAG, instructing it to generate detailed domain specifications based on the provided documents, for example “Using the retrieved passages from traffic regulations, convert the information into detailed and measurable safety specifications for [lane markings], ensuring each [lane marking] is defined with its precise meaning and characteristics.”

```
System Prompt: "Using the retrieved passages and traffic
       regulations, convert the information into detailed and
       measurable safety specifications for lane markings,
       ensuring each marking is defined with its precise
       meaning and characteristics."
# White Lane Line Markings:
1. Solid White Line
   Functional Requirements:
   - Urban Context: "The system shall detect solid white
   lane lines with a width of at least 15 cm when used
   to indicate the lane's shoulder or edge in city
   traffic."
   - Rural Context: "In rural settings, the system shall
   detect solid white lane lines with a minimum width
   of 10 cm, even if faded or partially obscured by
   road conditions."
   Non-Functional Requirements:
   - Reflectivity: "The system shall detect solid white
   lane lines with a minimum reflectivity of 70%
   during both day and night to ensure safe lane
   discipline."
   - Visibility: "The system shall identify solid white
   lane lines in low-visibility conditions (e.g., fog,
   heavy rain) from a distance of at least 50 meters."
   Contextual Adaptability:
   - Urban: Requires precise detection in high-density
   traffic environments, especially near
   intersections, merges, and exits.
   - Rural: Must ensure detection even when lines are
   faded or partially obstructed.
# Domain Specification continues...
```

2) *Knowledge Graph (KG) Construction*: A KG organizes and represents information about real-world entities and their relationships [17]. It is particularly useful for structuring ODDs extended specifications for better organization, reuse and dynamic reasoning. The adoption of KG structure in FODSE allowed engineers to easily query and reference detailed specifications, ensuring that all relevant conditions were captured with appropriate contextual variations.

To construct these structured representations, FODSE uses a Large Language Model to transform raw domain specifications into KG-compatible ontologies. During early-stage prototyping, we explored outputs from multiple LLMs—GPT-4o, DeepSeek-R1-Distill-Llama-70B, and Llama-3-8B—and selected GPT-4o [31], [46], [52] based on its consistency in producing structured, context-aware output across representative ODD attributes. This selection was guided by qualitative criteria such as format coherence and regulatory alignment; a formal benchmark was out of scope but remains future work. The GPT-4o’s familiarity with diverse ontology schemas (e.g., SCHEMA.ORG, FOAF, OWL) made it well-suited for organizing augmented ODD knowledge into structured KGs.

The requirement specifications generated in the previous step are structured into a KG. The engine employs the LLM, which uses a pre-trained ontology, namely SCHEMA.ORG, to organize the domain specifications into a structured format. This ensures that each attribute (e.g., lane markings) is systematically captured, allowing for efficient querying and contextual understanding.

```
{"@context": "https://schema.org", "@type": "Road",
  "name": "Solid White Line Markings",
  "description": "Indicates the edge of the roadway and
  separates traffic moving in the same direction.",
  "applicableRoadTypes": ["Highways", "Urban Streets",
  "Rural Settings"],
  "functionalRequirements": {"@type": "PropertyValue",
    "urban": {"name": "Solid White Line - Urban",
      "description": "Detect solid white lane lines with a
      width of at least 15 cm in city traffic."},
    "rural": {"name": "Solid White Line - Rural",
      "description": "Detect solid white lane lines with a
      minimum width of 10 cm, even if faded."}},
  "nonFunctionalRequirements": {"reflectivity": {"name":
    "Reflectivity", "value": "70%",
    "description": "Minimum reflectivity of 70% during both
    day and night."},
    "visibility": {"name": "Visibility", "value": "50
    meters", "description": "Identify lines from a
    distance of 50 meters in low-visibility
    conditions."}},
  "contextualAdaptability": {"urban": "Precise detection
  near intersections, merges, and exits.",
    "rural": "Detection of faded or partially obstructed
    lines."}}
```

¹<https://github.com/AI-EnabledSoftwareEngineering-AISE/FODSE>

```
// KG continues...
}
```

Additionally, in the following examples, you can see a simplified version of the prompt and the RAG's response to the user for the roadway user use case:

```
Query: "What are the different types of roadway users and
       their meanings?"
Documents: Sources include traffic regulations and
       transportation safety reports:
  - Pedestrians: "Includes individuals walking, using
    mobility aids, or jogging along roadways."
  - Bicyclists: "Cyclists operating standard bicycles or
    electric bikes in designated bike lanes or shared
    roadways."
  - Motorists: "Drivers of private vehicles, taxis, and
    rideshare vehicles."
  - Commercial Vehicle Operators: "Operators of trucks,
    delivery vans, and other freight vehicles."
  - Public Transit Users: "Passengers and operators of
    buses, trams, and other transit vehicles."
# Retrieved content continues...
```

```
System Prompt: "Using the retrieved passages, convert the
       information into structured and measurable safety
       specifications for roadway users, specifically
       focusing on bicyclist detection and safety in ADS."
# Bicyclist Safety Requirements:
1. Functional Requirements:
  - Urban Context: "The system shall detect cyclists in
    bike lanes with a minimum width of 1.5 meters and
    maintain a clearance of 1.5 meters when
    overtaking."
  - Suburban Context: "The system shall identify cyclists
    in shared roadways and maintain a 2-meter buffer
    zone when within 10 meters of an intersection."
  - Rural Context: "The system shall detect cyclists on
    road shoulders and provide a warning when
    overtaking if within a 100-meter detection range."
2. Non-Functional Requirements:
  - "The system shall detect cyclists at a minimum
    distance of 30 meters in low-light conditions with
    a 90% accuracy."
  - "The system shall provide a response time of less
    than 0.5 seconds to sudden cyclist movements within
    5 meters of the vehicle."
3. Contextual Adaptability:
  - Urban: High-density bike lanes, frequent
    intersections.
  - Suburban: Shared pathways, moderate cyclist presence.
  - Rural: Road shoulders, higher cyclist speeds.
# Domain Specification continues...
```

```
{"@context": "https://schema.org", "@type": "RoadwayUser",
  "name": "Roadway Users",
  "description": "Different types of users that interact
    with roadways, including pedestrians, bicyclists,
    motorists, and commercial vehicle operators.",
  "applicableUserTypes": ["Pedestrians", "Bicyclists",
    "Motorists", "Commercial Vehicles", "Public Transit
    Users"],
  "functionalRequirements": {"@type": "PropertyValue",
    "pedestrians": {"name": "Pedestrian Detection",
      "description": "Detect pedestrians crossing the
        roadway at speeds between 0.8 to 1.8 m/s and
        maintain a buffer zone of at least 3 meters."},
    "bicyclists": {"name": "Bicyclist Detection",
      "description": "Detect cyclists in bike lanes with a
        minimum width of 1.5 meters and maintain a clearance
        of 1.5 meters when overtaking."},
    "motorists": {"name": "Motorist Interaction",
      "description": "Detect sudden braking of vehicles
        and adjust vehicle speed to maintain a safe
        following distance of 3 seconds."},
```

```
"commercialVehicles": {"name": "Commercial Vehicle
  Interaction", "description": "Identify delivery
  vehicles and provide a safe passing buffer of at
  least 1.5 meters when lane-changing is not
  feasible."}
},
"nonFunctionalRequirements": {
  "detectionAccuracy": {"name": "Detection Accuracy",
    "value": "95%", "description": "The system shall
    maintain a 95% detection accuracy for all roadway
    users in low-light conditions."},
  "reactionTime": {"name": "Reaction Time", "value": "0.5
    seconds", "description": "The system shall respond
    to sudden movements of detected users within 0.5
    seconds."}
},
"contextualAdaptability": {
  "urban": "Frequent pedestrian crossings, high bicycle
    density.",
  "suburban": "Mixed-use pathways, moderate motorist and
    cyclist interaction.",
  "rural": "Road shoulders, occasional commercial vehicle
    presence."
}
// KG continues...
}
```

Traditionally, this process would involve manual reviews of road condition data and updates to the requirements. However, using FODSE framework, the process became more efficient. The RAG engine in FODSE automatically extracted relevant information from domain-specific sources, such as road condition reports and urban planning documents and augmented its prior extensive knowledge to synthesize and transform ODD attributes—such as lane marking types (continuous, dashed, double lines), lighting conditions (daytime, nighttime), and roadway features (curves, intersections)—into domain-specific requirements.

B. Dataset Evaluation Engine

The dataset evaluation engine includes Specification Mapping (Figure 1-3) and Report Mapping (Figure 1-4) components.

1) *Specification Mapping*: This component evaluates a given dataset based on the defined classes of ODD attributes established in the previous step. Specifically, it classifies images according to different types of an ODD attribute (e.g., lane markings), such as continuous white lines or dashed yellow lines, which were previously defined. The primary objective of this engine is to identify underrepresented or missing ODD attribute variants in a dataset, ensuring that dataset gaps are systematically detected and addressed.

To operationalize this mapping, the engine uses textual prompts generated through Specification Augmentation, which translates structured ODD attributes into model-compatible natural language phrases. These prompts represent lane markings and roadway user variants and serve as inputs to the multimodal classification process. This mechanism enables dataset coverage analysis even when only partial ground-truth annotations are available, allowing a consistent pipeline from semantic specifications to model-based evaluation.

To process these prompts, FODSE leverages multimodal models (MM), which can process multiple types of input modalities—such as text and images. These models enable

the identification of ODD attributes within image datasets by mapping textual specifications of ODD attributes to their corresponding visual representations. In our implementation, we employ text-vision multimodal models, including CLIP and BLIP2, to analyze the presence of ODD attributes across both natural language and image modalities. This allows for a structured comparison between expected domain specifications and real-world dataset coverage.

To improve the classification accuracy of MM models and optimize their ability to recognize ODD attributes, we integrate *prompt learning*. Prompt learning is a technique that fine-tunes how an AI model interprets and responds to input prompts without requiring additional training data. Instead of traditional retraining, prompt learning refines the way the model processes prompts, making it more context-aware and better suited to the specific classification task [54].

For this purpose, we employ *PromptSRC*, a framework designed for prompt learning with *Self-Regulating Constraints* [19]. Unlike manually crafted text-based prompts, PromptSRC optimizes prompt structures by refining their numerical embeddings. These embeddings are iteratively tested and adjusted based on model performance, ensuring that the prompts guide the model toward improved classification accuracy. If the model fails to achieve the desired accuracy, the embeddings are automatically refined and re-evaluated until an optimal prompt structure is identified. This process enhances the model’s ability to interpret and classify ODD attributes more effectively while preserving its pre-trained knowledge. In our case, the *initial prompt* passed to the model consisted of the text:

“A photo of a [ODD attribute class: continuous white line]”

paired with an image containing an image of this class. The model then converted this text-image pair into embeddings and iteratively refined these embeddings to improve classification accuracy. During the training phase, the model was trained for 20 epochs, using evaluation sets to determine the most effective prompt variation. The best-performing model was selected based on its classification F1-score.

It is important to note that these optimized prompt embeddings exist in a numerical vector space and do not have direct human interpretability. Unlike textual prompts that can be explicitly written and modified, prompt embeddings are latent representations that the model internally refines. Therefore, providing explicit examples of the optimized embeddings is not meaningful in a human-readable format. Instead, we evaluate their effectiveness based on the model’s classification performance improvements.

Since our task involves multilabel classification, where each image may contain multiple ODD attributes, we modified PromptSRC to support multiple true labels per image. This required modifying the loss function, replacing the traditional cross-entropy loss (used for single-label classification) with a multilabel loss function. This adjustment allowed the model to predict multiple ODD classes simultaneously while treating each label independently. Additionally, we updated the model’s

output layer to generate class-wise probabilities, ensuring accurate classification even in images where multiple ODD attributes coexist. The implementation code for this process is publicly available in our repository.

The integration of prompt learning in FODSE significantly improved the identification and classification of ODD attributes within image datasets. By optimizing the way ODD specifications are represented within multimodal models, our approach enhances dataset evaluation without requiring additional labeled training data. For example, this technique allowed the model to correctly differentiate between lane marking variations—such as solid and dashed lines—without requiring explicit exposure to additional training images. This demonstrates the potential of prompt learning in domain-specific dataset assessments, improving dataset completeness and overall ADS model training.

2) *Report Generation*: This component processes the outputs of the MM models and generates a structured report, including statistics summarizing the dataset’s coverage of ODD attributes framed as domain specifications, such as the percentage of images containing certain lane markings or traffic scenarios. It provides valuable insights into the dataset’s characteristics and any gaps that may need attention.

In this phase, the extracted domain specifications act as a benchmark against which the dataset is evaluated. In the previous step, by leveraging the MM models, the approach identifies and classifies the visual elements within the dataset (images) according to these specifications. This step then quantifies the presence, distribution, and diversity of these attributes to evaluate the dataset’s overall quality and suitability for training autonomous systems.

V. EVALUATION

To assess the effectiveness of the proposed framework, we evaluated its two core components: the Specification Augmentation Engine and the Dataset Evaluation Engine, in two use cases of ODD categories: *lane markings* and *roadway users*. We evaluate how well the former engine translates ODDs into requirements specifications and how effectively the latter engine uses these specifications to accurately assess their presence in an image-based dataset.

A. Assessment of Specification Augmentation Engine

Given the ODD challenges discussed in Section II, including their lack of specificity, ambiguity, and contextual differences, we evaluate the first engine on how well its output address these challenges by transforming ODD classes into structured domain specifications to support downstream analysis.

We extracted a comprehensive list of domain specifications by applying the FODSE framework to two ODD categories of *lane markings* and *roadway users*. For lane markings, FODSE identified 12 classes, including White Lane Line Markings, Yellow Center Line Pavement Markings, Edge Line Pavement Markings, Crosswalk Markings, Stop and Yield Lines, Chevron and Diagonal Crosshatch Markings, Preferential Lane

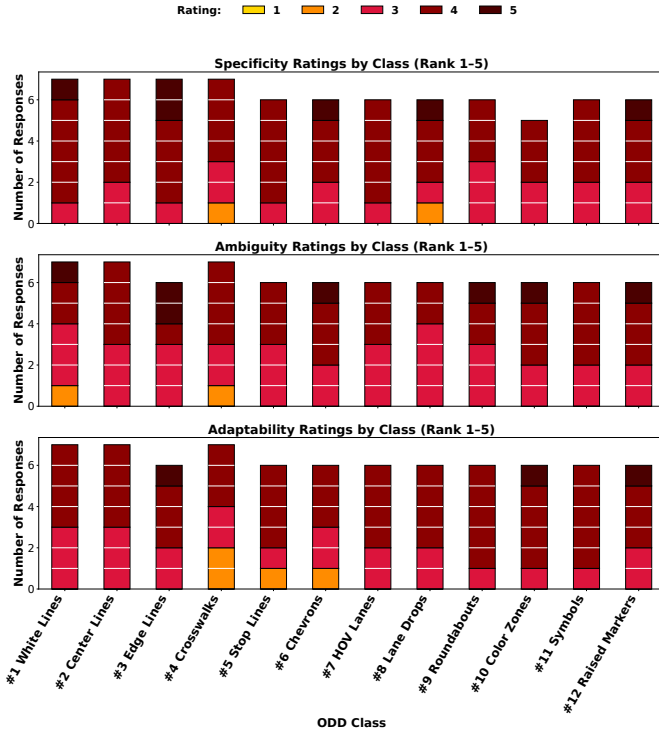


Fig. 2. Stacked distribution of expert Likert-scale ratings (1–5) for Specificity, Ambiguity, and Adaptability across **Lane Marking** categories. Each horizontal panel shares a consistent layout and class alignment.

Markings, Lane Drop Markings, Roundabout Markings, Colored Pavements, Word, Symbol, and Arrow Markings, and Raised Pavement Markers. For roadway users, FODSE delineated 15 classes, including Pedestrians, Bicyclists, Micromobility Users (electric scooters, e-bikes, skateboarders), Individuals with Disabilities, Motorists (including commercial vehicles), Motorcyclists, Public Transit Users, Agricultural Equipment Operators, Construction and Maintenance Workers, Equestrians, Horse-drawn Vehicle Operators, Freight and Delivery Vehicles, Power Wheelchairs and Mobility Scooters, Inline Skaters, and Cargo Bicyclists.

FODSE then associated each condition with a comprehensive list of functional and non-functional requirements specifications. For lane markings FODSE extracted 25 functional, 22 non-functional, and 25 contextual adaptability specifications. Contextual Adaptability in ODD domain specification refers to varying environmental and operational conditions, in which the specified requirements may vary in detail, and therefore, require the system to adjust its behavior and detection capabilities accordingly. For example, the requirement for detecting solid white lane markings may differ between *urban* and *rural* settings: in urban areas, the lines must be at least 15 cm wide, whereas, in rural areas, they can be as narrow as 10 cm, according to the road standards. Thus, the system is required to adapt to these context-specific variations to ensure accurate detection and compliance across different environments. For roadway users, FODSE identified 51 functional, 30 non-

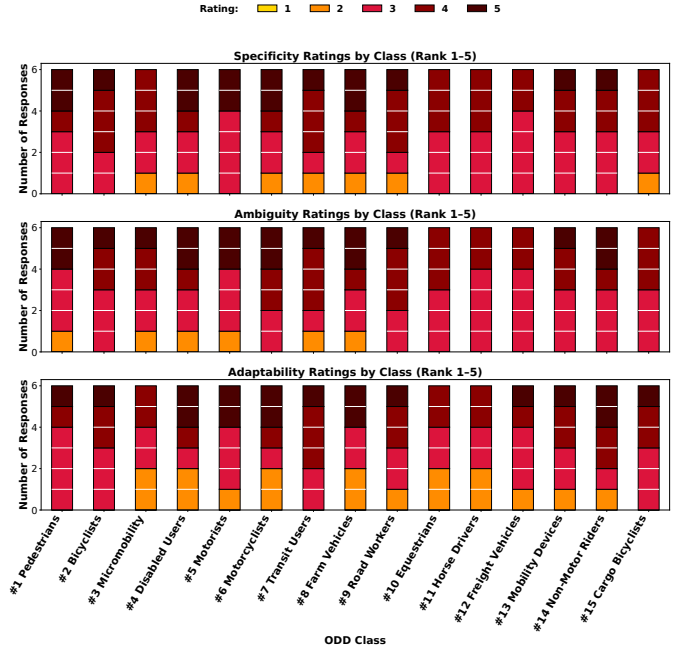


Fig. 3. Stacked distribution of expert Likert-scale ratings (1–5) for Specificity, Ambiguity, and Adaptability across **Roadway Users** categories. Each horizontal panel shares a consistent layout and class alignment.

functional, and 39 contextual adaptability specifications. The complete list of these specifications is in our repository.

To evaluate our approach, we presented 192 requirements specifications to a panel of seven experts, consisting of researchers and developers from General Motors R&D with diverse experience levels in autonomous driving R&D. The experts were provided with an online evaluation form (available in our repository), where they were initially asked to self-assess their software engineering experience in the automotive industry as advanced (high), skilled (medium), or beginner (low). Among them, four identified as skilled, two as advanced, and one as a beginner.

Additionally, the form included a question assessing their familiarity with lane marking detection and pedestrian detection, which rely on the roadway users ODD attribute in ADS. Participants were asked to rank their familiarity as familiar, somewhat familiar, or not familiar. Five participants rated themselves as familiar, one as not familiar, and one as somewhat familiar. This balanced distribution of expertise ensured that the evaluation incorporated both experienced and less familiar perspectives, providing a more comprehensive assessment of FODSE’s effectiveness.

The participating experts, specializing in software engineering for automotive safety, were independent of this study and provided objective assessments, serving as a benchmark for evaluating our approach’s effectiveness.

We then asked the experts to evaluate the engine’s outputs according to three key metrics: **Specificity**, **Ambiguity Resolution**, and **Contextual Adaptability**. *Specificity* measures how detailed and precise the functional and non-functional

requirements are. We further clarified that the *ambiguity resolution* aims to assess how well the requirements reduce uncertainty and clarify expectations, while *contextual adaptability* examines how well the specification addresses different operational contexts (e.g., urban vs. rural environments). We asked participants to rank each requirement based on the three metrics of specificity, ambiguity and adaptability on a scale from 1 to 5, where 1 indicates “not specific/ambiguous/not adaptable” and 5 indicates “specific/clear/adaptable.”

To make expert review feasible, we selected a representative specification per class, resulting in 12 lane marking and 15 roadway user specifications evaluated across three dimensions. Figures 2 and 3 present the full distribution of Likert-scale ratings (1–5) for Specificity, Ambiguity, and Contextual Adaptability across all evaluated classes. These stacked bar charts visualize 252 expert responses for Lane Markings (7 experts $\times$ 12 classes $\times$ 3 metrics) and 270 for Roadway Users (6 experts $\times$ 15 classes $\times$ 3 metrics), revealing detailed rating patterns beyond summary statistics. Lane Marking specifications received consistently strong scores, with most ratings concentrated between 3 and 5, and very few low-end responses, indicating that these specifications were generally viewed as clear, specific, and context-aware. Roadway User specifications showed similar trends on Specificity and Ambiguity, while Adaptability revealed a broader spread, including 17 ratings of 2, highlighting reduced expert confidence in generalization across varying driving scenarios.

These distributions enable a safety-aware interpretation of specification quality: even a minority of low scores can surface context-specific vulnerabilities. While many responses clustered around the midpoint (3), this reflects the inherent complexity of defining adaptable specifications for open-world concepts such as “Cargo Bicyclists” or “Road Workers.” Rather than aiming for exhaustive precision, our framework supports partial and evolving specifications, enabling mid- or low-rated cases to be iteratively revised and re-evaluated through targeted improvement.

While the assessment results indicate that FODSE performed well in transforming ODD classes into more precise, explicit and structured domain specifications with reduced ambiguity, expert feedback highlighted the need for future refinements in incorporating environmental factors such as weather and visibility. After further discussions with the experts, their insights highlighted opportunities for future enhancement of applicability of generated specifications through the incorporation of regional weather information, ensuring that the specifications better support real-world automotive safety and autonomous driving development. By integrating localized climate data, the system can account for variations in road conditions, visibility, and traction influenced by factors such as heavy rain, snow, fog, or extreme temperatures. This would enable the generation of more context-aware requirements that adapt to diverse environmental conditions, improving the robustness and reliability of autonomous driving systems.

These expert evaluations not only quantify specification quality but also reinforce FODSE’s alignment with core RE

principles. Visualizing full Likert-scale distributions—rather than relying on averages—reveals both central trends and edge-case disagreements critical in safety-oriented RE contexts. The structured specifications generated here served as operational artifacts for subsequent dataset evaluation, demonstrating how partial and evolving requirements can directly support data-driven analysis. This positions FODSE as a practical RE tool that bridges high-level domain assumptions and dataset-level implementation—setting the stage for the next component’s dataset coverage assessment.

B. Assessment of Dataset Evaluation Engine

We assess this engine by evaluating its ability to accurately characterize the defined ODD classes within image datasets. A thorough evaluation of the dataset against the extended classes of ODD classes helps identify underrepresented ODD variants, enabling a systematic approach to augmenting the dataset with missing or insufficiently represented conditions. By addressing these gaps, we can enhance dataset diversity, leading to more comprehensive model training and improved accuracy in autonomous system performance. This process ensures that the trained models are better equipped to handle real-world variability, ultimately contributing to more robust and reliable autonomous driving solutions.

1) *Experimental Setup:* For evaluation, we initially prepared 27 specification prompts—12 for lane markings and 15 for roadway users—based on the output of the Specification Augmentation module. However, only 6 lane types and 7 user classes had corresponding ground truth annotations across our datasets. As a result, the final evaluation was limited to 13 ODD classes for which both specifications and labels were available.

We used two state-of-the-art multimodal models, namely CLIP [34] and BLIP2 [24], in their original forms, while in FODSE its prompt learning framework is applied to the original CLIP model as earlier discussed.

For BLIP2, we employed it as a Visual Question Answering (VQA) model, posing a simple binary question as, “Do you see a [ODD class] in this image?” The model’s yes/no answers were then compared to the ground truth, presented by the labels assigned to the images in the dataset. For CLIP, which outputs probabilities for the presence of a label in images, we considered probabilities exceeding 0.50 to be present.

In FODSE, prompts were optimized using the modified PromptSRC framework as previously explained. We retrained each model for 20 epochs, with the dataset split into 50% for training, 20% for validation, and 30% for testing. After identifying the best-performing prompts, the test data was evaluated using the same process as the standard CLIP model to determine labels.

In the case of lane markings, we utilized the TuSimple dataset [49], containing 6,408 road images captured on U.S. highways under various conditions. Of these, only 3,626 images include ground truth annotations for six lane-marking classes out of our 12 identified lane-marking classes, including Continuous Yellow, Continuous White, Dashed, Double

TABLE II
EXPERIMENTAL RESULTS FOR LANE MARKINGS

	Accuracy			Precision			F1-Score		
	BLIP	CLIP	FODSE	BLIP	CLIP	FODSE	BLIP	CLIP	FODSE
Continuous Yellow	62.0	38.2	85.9	1.00	38.2	90.0	3.0	55.3	79.5
Continuous White	52.0	47.7	85.6	0.50	47.7	84.2	4.0	64.6	85.0
Dashed	49.0	52.1	96.8	0.67	52.1	98.2	5.0	68.5	96.9
Double Dashed	89.0	10.8	90.8	0.0	10.8	73.0	0.0	19.4	35.1
Botts' dots	52.0	47.7	89.1	0.0	47.7	99.8	0.0	64.6	87.1
Double Continuous	85.0	15.2	90.3	0.0	15.2	70.5	0.0	26.3	66.2
Average	65.0	35.3	89.8	0.36	35.3	85.9	2.0	49.8	75.0

TABLE III
EXPERIMENTAL RESULTS FOR ROADWAY USERS

	Accuracy			Precision			F1-Score		
	BLIP	CLIP	FODSE	BLIP	CLIP	FODSE	BLIP	CLIP	FODSE
Bicycle	93.0	7.4	92.2	67.0	7.4	40.0	6.0	13.8	14.8
Bus	85.0	15.8	85.0	67.0	15.8	53.5	8.0	27.2	44.4
Car	21.0	99.4	95.6	100.0	99.4	99.5	34.0	99.7	97.8
Motorcycle	94.0	6.1	94.5	40.0	6.1	66.7	12.0	11.5	29.0
Pedestrian	61.0	39.0	70.9	50.0	39.0	60.5	1.0	56.1	66.3
Rider	90.0	7.4	93.6	18.0	7.4	84.6	12.0	13.8	27.8
Truck	70.0	29.1	70.6	42.0	29.1	49.4	9.0	45.0	47.7
Average	73.0	29.2	86.1	55.0	29.2	64.9	12.0	38.2	46.8

Dashed, Bott's Dots, and Double Continuous. While this dataset limitation highlighted the value of our approach in verifying the completeness of collected datasets, it impacted our evaluation, forcing us to assess only six of the twelve lane-marking classes.

For the roadway user detection use case, we utilized a 10k subset of the BDD100K dataset [50]. Of these, only 2,956 images included ground truth labels, covering seven of our 15 identified roadway user classes, including Bicycles, Buses, Cars, Motorcycles, Pedestrians, Riders, and Trucks. We only evaluated the classes that had existing ground truth labels.

2) *Experimental Results and Analysis*: Tables II and III summarize the performance of the models on lane markings and roadway users, respectively. Across both tasks, CLIP without prompt learning exhibited unreliable behavior by predicting the existence of every class in nearly every image, leading to high false positives and misleading results. This resulted in precision and accuracy being inflated due to the absence of true negatives or false negatives, creating identical precision and accuracy values, which do not reflect actual performance.

BLIP performed poorly across both lane markings and roadway users, particularly in terms of F1-score, which remained near zero for most classes in both tasks. This indicates that BLIP struggled to differentiate between the various lane marking types and roadway user categories. A notable exception was the "car" class in the roadway user task, where BLIP achieved perfect precision (100%) but suffered from high false negatives, reducing overall accuracy.

In contrast, the prompt-learned version of CLIP in FODSE showed significant improvements in both precision and F1 scores across all classes in both tasks. For instance, in the lane

marking task, FODSE achieved an F1 score of 96.8 for dashed lines and 85.0 for continuous white lines. Similarly, in the roadway user task, FODSE performed well on key classes such as "car" (F1-score: 97.8) and "pedestrian" (F1-score: 66.3). This improvement demonstrates that prompt learning enhances the model's ability to correctly classify domain-specific ODD classes across a wide range of conditions, compared to base multimodal models.

The experimental results highlight the limitations of existing multimodal models in accurately identifying ODD classes and the effectiveness of prompt learning in enhancing their performance. CLIP, in its original form, exhibited unreliable behavior, over-predicting the presence of classes and inflating precision and accuracy metrics due to the lack of true negatives. Similarly, BLIP2 struggled to differentiate between various lane marking types and roadway user categories, particularly demonstrating near-zero F1-scores across most classes. These findings emphasize the inherent challenges in using general-purpose vision-language models for domain-specific tasks without adaptation.

In contrast, our framework enabled the model to more accurately detect the presence of ODD classes and distinguish between different class types, resulting in higher precision and F1-scores across all evaluated categories.

These results reinforce the effectiveness of FODSE's two-stage approach, not only in producing structured and reviewable ODD specifications, but also in enabling their practical utilization. The structured specifications guided dataset gap detection and significantly improved classification accuracy over baseline models, demonstrating how RE-style artifacts can operationally support dataset validation and model alignment—even when full class coverage is unavailable. This, in

turn, supports more effective model training and contributes to the overall reliability of perception models in autonomous driving systems.

VI. THREATS TO VALIDITY

The knowledge graph and report generation components were primarily included for representation purposes to help our partner engineers interpret the findings from FODSE’s core engines—specification augmentation and dataset evaluation. Since these components were not essential to the functionality of our approach, they were not evaluated. However, we acknowledge that alternative representation methods could potentially offer better interpretability. Despite this, the overall effectiveness of FODSE remains robust and impactful, as its core mechanisms were the primary focus of evaluation.

Reliance on RAG for framing ODDs as structured domain specifications introduces potential risks, including dependence on the quality and potential biases of retrieved data. If the retrieval process is skewed toward certain perspectives or outdated sources, it may lead to incomplete or inaccurate specifications.

Another challenge is contextual misalignment, where the model might misinterpret the operational domain and generate misaligned requirements. We mitigated these risks through human-in-the-loop oversight, ensuring continuous monitoring and refinement of the generated outputs. However, this process remains semi-automated, and future work could explore improved retrieval filtering techniques or fine-tuning domain-adapted language models for more precise specification extraction.

Another limitation stems from our reliance on publicly available regulatory and industry documents to contextualize the model’s outputs. While we carefully selected high-quality sources to maximize relevance, there is always a possibility of noise, inconsistencies, or missing critical nuances in the retrieved content. Some proprietary or industry-specific documents, which were inaccessible, could provide more comprehensive domain coverage. Future extensions of this work could incorporate privileged access to proprietary datasets to enhance specification accuracy.

Dataset limitations and class imbalance also present validity concerns. In TuSimple and BDD100K, only a fraction of the ODD attributes framed as domain specifications contained ground truth annotations, limiting our evaluation to those attributes. Additionally, some attributes were over-represented, introducing potential biases in model performance metrics, particularly accuracy. While prompt learning improved classification accuracy, the underlying multi-modal AI models (CLIP, BLIP2) were originally trained on broad, general-purpose data and may not fully capture the intricacies of domain-specific ODD attributes. To address the limitations imposed by partial annotations and class imbalance, future work will explore annotation-free validation strategies. These include (i) open-vocabulary object detectors such as Grounding DINO [25] and OWL-ViT [28] for pseudo-labeling, (ii) explanation-based alignment checks (e.g., Grad-CAM [39]),

and (iii) distributional coverage metrics to assess semantic alignment with ODD specifications. Such methods would enable scalable and specification-driven validation of dataset compliance, even in the absence of complete ground truth.

Furthermore, we did not conduct extensive generalization experiments with the prompt-learned models due to ground truth constraints, narrowing the scope of our findings. Simplifying complex domain specifications into discrete classification tasks was necessary for our evaluation. However, future research should explore more nuanced validation strategies, including continuous learning approaches, more diverse datasets, and multimodal AI models trained specifically for ODD assessment. This would further enhance FODSE’s adaptability and effectiveness in refining ODD specifications for autonomous driving applications.

VII. CONCLUSION AND FUTURE WORK

This paper presents the FODSE framework, a practical solution for aligning datasets with well-defined ODD specifications to enhance ADS robustness. By leveraging multimodal AI models, prompt learning, and RAG, FODSE first translates ODD attributes into structured specifications and then validates their presence in datasets. By framing ODD attributes as explicit domain-level requirements and providing mechanisms to assess their representation in data, FODSE enables core Requirements Engineering (RE) practices—such as traceability, ambiguity resolution, and validation—to be applied directly to dataset analysis. This integration supports the practical development of verifiable and accountable autonomous driving systems. Evaluation results show its effectiveness and point to areas for future improvement.

Future work will enhance FODSE’s adaptability and precision across different contexts. A key focus is integrating regional factors like weather and visibility to refine specification generation. Accounting for dynamic environmental variations—such as rain, fog, snow, and extreme temperatures—will improve ADS robustness.

Another priority is extending prompt learning to infrastructure elements, road conditions, and traffic scenarios, while optimizing for low-resource environments to enhance adaptability across geographies and regulatory landscapes.

Lastly, improving dataset augmentation will address limitations identified during evaluation. FODSE can use synthetic data generation and domain adaptation techniques to enrich underrepresented ODD attributes, ensuring better generalization across diverse conditions.

ACKNOWLEDGMENTS

We thank the NIU oneAPI Center of Excellence and Intel’s AI Tiber Cloud team for their support and resources. Special thanks to Kittur Ganesh (Intel) for his valuable guidance throughout this project.

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